PROGRAM TO SEARCH AN ELEMENT IN ARRAY USING BINARY SEARCH

AIM:

To write array program using binary concepts

ALGORITHM:

STEP 1: start

STEP 2: declare no of elements (n) elements to search (key)

STEP 3: initialization array with n(arr[n])

STEP 4: i=0

STEP 5: get arr[i]

STEP 6: i++

STEP 7: if i<n goto step 5,else continue

STEP 8: read element to search(key)

STEP 9: result = binary search(arr,n,key)

STEP 10: if (result!=1)

STEP 11: print element found at position(result)else

print element not found

STEP 12: stop

ALGORITHM FOR BINARY SEARCH:

STEP 1: int low=o, high=n-1

STEP 2: if low(<=high)

STEP 3: mid=(low+high)/2

STEP 4: if (arr[mid]==key)

return mid

STEP 5: else if (arr[mid]<key)

low=mid+1

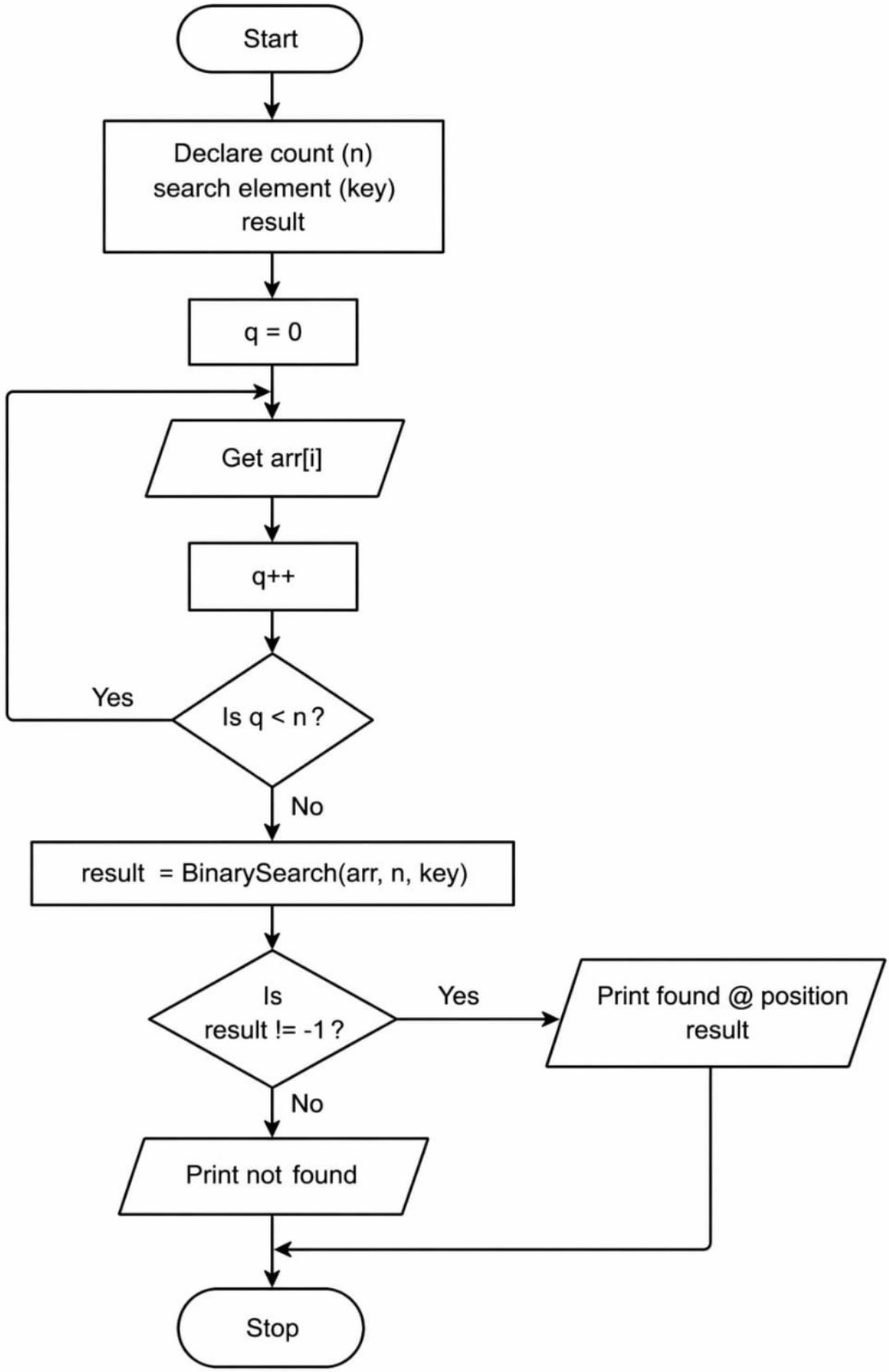
else

high+mid-1

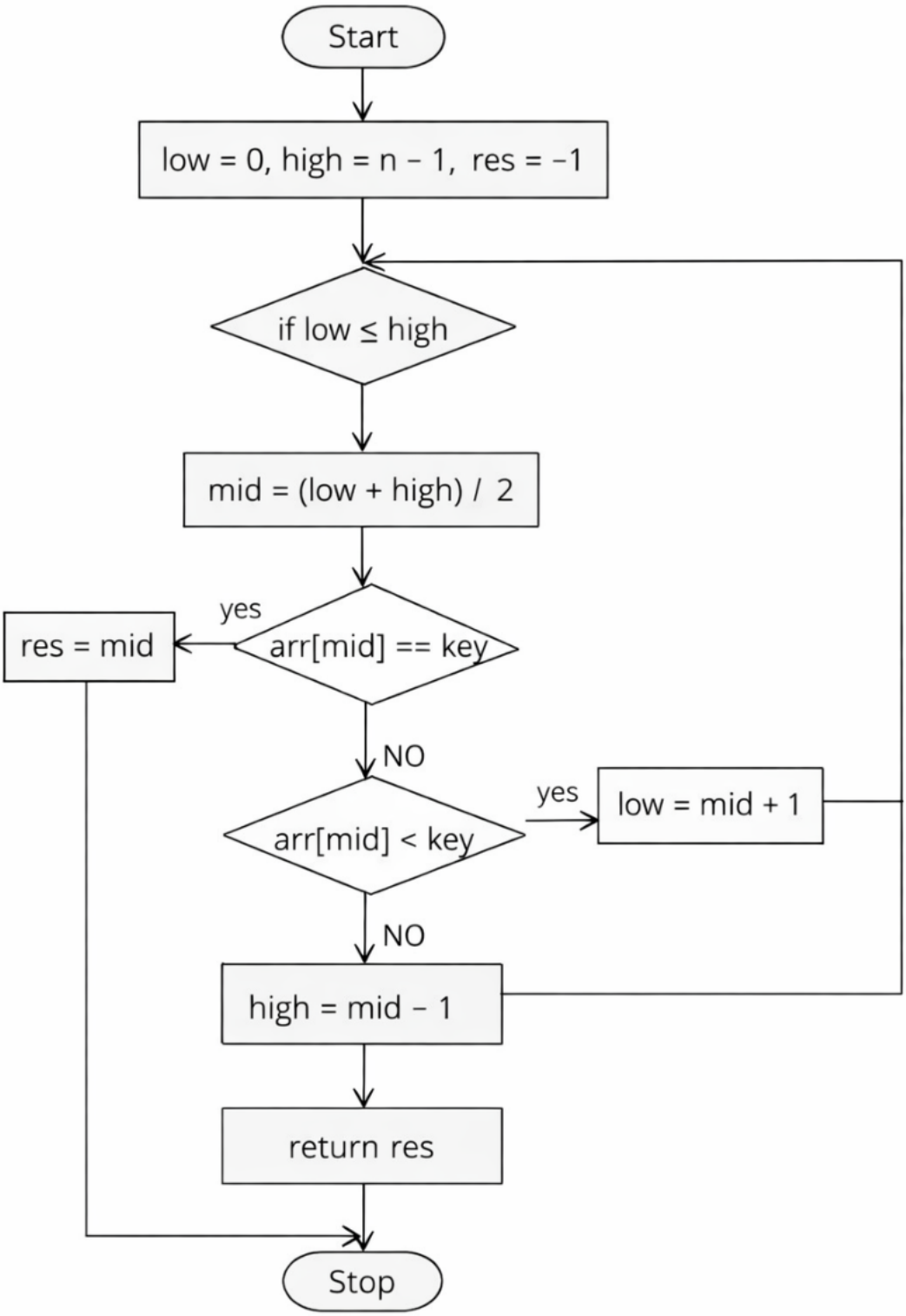
STEP 6: goto step 2

STEP 7: return-1

FLOWCHART :


Start
Declare count (n)
search element (key)
result
q = 0
Get arr[i]
q++
Yes
Is q < n ?
No
result = BinarySearch(arr, n, key)
Is
result != -1 ?
Yes
Print found @ position
result
No
Print not found
Stop

ALGORITHM FOR BINARY SEARCH :


Start
low = 0, high = n − 1, res = −1
if low ≤ high
mid = (low + high) / 2
yes
res = mid
arr[mid] == key
NO
yes
low = mid + 1
arr[mid] < key
NO
high = mid − 1
return res
Stop

CODING :

```
#include<stdio.h>
int binarysearch(int arr[],int n, int key)

{
int
low=0,high=n-1,mid;

while(low<=high)

{

mid=(low+high)/2;

if (arr[mid]==key)
return mid;
//element found
else if(arr[mid]<key)
low=mid+1;
//search right half
else
high=mid-1;
//search left half
}
return 1;
//element not found
}

int main()

{
int n,key,result;

printf("enter number of element:");
```

```
scanf("%d",&n);

int arr[n];

printf("enter %d sorted element:\n",n);
int i;
for(i=0;i<n;i++){

scanf("%d",&arr[i]);
```

```
}

printf("enter element to search:");

scanf("%d",&key);

result = binarysearch(arr, n, key);

if(result!=-1)

printf("element found at index %d \n",result);
else
printf("element not found \n");
return 0;
}
```

OUTPUT :

```
enter no of element                    : 5
enter 5 sorted element
 11
 12
 13
 14
 15
enter element to search : 14
enter found at index : 3
```

RESULT :

Thus the program compiled and executed successfully with verified output